

FEDERAL BOARD OF INTERMEDIATE & SECONDARY EDUCATION

Detailed Marking Report — Physics Exam-14

Seemab · Class 9 (Science Group) · Revision Test, Units 1 to 6 and Unit 9 · Attempted 21 August 2026

TOTAL SCORE 37 / 43	PERCENTAGE 86.0%	GRADE A	SECTION A 20 / 24	SECTION B 9 / 9	SECTION C 8 / 10
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How this paper was marked. As with the Chemistry paper, Seemab listed her MCQ answers rather than filling the grid, and three of them (Q5, Q9, Q10) carry a scribbled-out first attempt with a clear final letter beside it. Every answer was re-read from the scan at 400 DPI, the answer key was re-extracted independently from the published PDF, and the printed option order for each of her four wrong answers was checked against the paper. **All four are sound, on-SLO questions and no question needed voiding.**

SECTION A — Multiple Choice (20 / 24)

#	Topic	Hers	Key	Mark	Comment and source in her textbook
1	Screw gauge least count	C	C	1	Correct. U1 §1.7.3, SLO P-09-A-19.
2	Scientific notation of 0.00085	A	A	1	Correct. U1 §1.4, SLO A-05.
3	Which is a vector	D	D	1	Correct, momentum. SLO A-09.
4	Prefix for 10^{-9}	C	C	1	Correct, nano. Table 1.3.
5	Area under a speed-time graph	A	B	0	She chose "acceleration". The area gives distance travelled; the gradient gives acceleration. SLOs B-12/B-13. This same confusion cost her 2 more marks in Section C , see below.
6	Free-fall acceleration	A	A	1	Correct, 9.8 m/s^2 . SLO B-10.
7	Non-uniform acceleration	D	D	1	Correct. SLO B-08.
8	Average speed, 96 m in 6 s	B	B	1	Correct, 16 m/s. SLO B-04.
9	Newton's second law	C	C	1	Correct after crossing out a first attempt. SLO B-30.
10	SI unit of impulse	D	D	1	Correct, N s, again after a correction. SLO B-40.
11	Limitations of Newton's laws	A	A	1	Correct, the atomic scale. U3 §3.2.4, SLO B-33. First time this was set as an MCQ.
12	Instrument for measuring mass	A	B	0	She chose "force meter". A force meter measures weight ; mass is measured with an electronic balance . SLOs B-21/B-22.
13	Centre of gravity	C	C	1	Correct. U4 §4.3, SLO B-47.
14	Orbital speed formula	A	A	1	Correct, $v = 2\pi r/T$. U4 §4.8.1, SLO F-01.
15	Like parallel forces	D	D	1	Correct. U4 §4.2, SLO B-43.
16	Principle of moments	B	B	1	Correct. U4 §4.4.2, SLO B-45.
17	Unit of spring constant	A	A	1	Correct, N/m. SLO B-56.
18	Limit of proportionality	C	C	1	Correct. SLO B-58.
19	Instrument for atmospheric pressure	B	B	1	Correct, barometer. U5 §5.3.1, SLO B-83.
20	Mercury column at sea level	D	D	1	Correct, 76 cm. U5 §5.3.1.
21	Non-renewable resource	D	B	0	She chose "geothermal". The non-renewable option was coal ; geothermal is renewable. U6 §6.3, SLO B-70. She defined both terms correctly and named coal herself in Section C , so this is a slip rather than a gap.
22	One watt	C	C	1	Correct, 1 joule per second. U6 §6.4, SLO B-73.
23	Perpetual motion machine	C	A	0	She chose "moments". A perpetual motion machine is impossible because it violates the conservation of energy . SLOs B-68/B-69.
24	Falsifiability	D	D	1	Correct. U9, SLO P-10-G-07.

SECTION B — Short Questions (9 / 9)

Q.2(i) Vernier caliper 3 / 3

(a) Least count = $1 \div 50 = 0.02$ mm, then converted to **0.002 cm**. Correct. 1

(b) She laid the reading out properly: main scale reading 6.7 cm, coinciding division 25th, vernier scale reading = $25 \times 0.002 = 0.05$ cm, total length = $6.7 + 0.05 = \mathbf{6.75\text{ cm}}$. Correct. 2

Small habit to fix: her final line reads "6.75" with no unit. The unit appears earlier in her working so the mark stands, but a board examiner can be strict about the final answer carrying its unit.

Q.2(ii) Centre of mass and centre of gravity 3 / 3

(a) She set the two out as a clean two-column comparison: centre of mass as "the point where the whole mass of body can be considered to be concentrated", centre of gravity as "the point where the whole weight of body can be considered to be act". Both correct. 2

(b) This was the best-reasoned answer on either paper:

"They aren't the same when gravitational field is non-uniform (it's strength or direction varies across the body). In a uniform gravitational field, the centre of mass and centre of gravity are the same point."

That is a sharper statement of the condition than the mark scheme itself asked for, which expected only "a very large body where g is not the same at all parts". 1

Q.2(iii) Barometer with a different liquid 3 / 3

(a) She started from $P = \rho gh$, set $\rho_1 g h_1 = \rho_2 g h_2$, cancelled g , and solved $13600 \times 76 = 3400h$ to get $h = 4 \times 76 = \mathbf{304\text{ cm}}$. That is the mark scheme's 3.04 m expressed in centimetres, which is entirely consistent because she kept the 76 in cm throughout. Correct. 2

(b) "for same pressure, $h \propto 1/\rho$ ". Correct, and concisely put. 1

SECTION C — Long Questions (8 / 10)**Q.3(i) Speed-time graph** 3 / 5

(a) **Describing the three stages** 1 / 1 — she labelled them AB, BC and CD rather than A, B and C, but the physics is right:

"AB :- acceleration BC :- zero acceleration rest/constant CD : deacceleration"

(b) **Acceleration during stage A** 0 / 2 — this is the one real error on the paper. She wrote:

"Triangle :- $\frac{1}{2}(bh) \rightarrow \frac{1}{2}(10 \times 14) \rightarrow \frac{1}{2}(140) = 70\text{ m/s}^2$ "

She calculated the **area** of the first triangle and then labelled that area as an acceleration in m/s^2 .

Acceleration is the **gradient**, $a = (v - u) \div t = (14 - 0) \div 10 = \mathbf{1.4\text{ m/s}^2}$. No gradient formula appears anywhere in her working, so no method mark can be given.

(c) **Total distance** 2 / 2 — and here she used the area method correctly and completely: $\frac{1}{2}(10 \times 14) + (20 \times 14) + \frac{1}{2}(10 \times 14) = 70 + 280 + 70 = \mathbf{420\text{ m}}$. Exactly right.

The one misconception worth all the attention. Look at these three answers together: in MCQ 5 she said the area under a speed-time graph gives **acceleration**; in Q.3(i)(b) she computed an **area** and called it an acceleration; and in Q.3(i)(c) she used an area to find **distance**, correctly. She is not confused about how to find an area. She has not separated **gradient** from **area**. One idea, fixed in a single sitting, is worth **3 marks** on this paper: **the slope of a speed-time graph is the acceleration, the area underneath it is the distance.**

Q.3(ii) Energy resources and power 5 / 5

(a) Renewable as "energy resources that are naturally replenished and can be used repeatedly", with solar and wind; non-renewable as "energy resources that are finite and will eventually run out", with coal and natural gas. Two correct examples of each. 2

(b) Hydroelectric advantage and disadvantage, both correct:

"It produces electricity without burning fuel, so there is very little air pollution during operation." / "Building dams can flood large areas of land and affect ecosystems."

1

(c) The conveyor-belt numerical, fully correct: $d = vt = 2 \times 90 = 180\text{ m}$, $W = Fd = 350 \times 180 = \mathbf{63,000\text{ J}}$, then $P = W/t = 63,000 \div 90 = \mathbf{700\text{ W}}$. Formula, substitution, answer and units all present. 2

What to revise, in priority order

- **Gradient versus area on a speed-time graph.** Worth 3 marks on this paper alone. Slope = acceleration. Area underneath = distance. Nothing else on this paper comes close in value.
- **Which instrument measures what.** An electronic balance measures **mass**; a force meter or spring balance measures **weight**.
- **Conservation of energy as the reason a perpetual motion machine cannot work.** It would create energy from nothing.
- **Renewable list.** Geothermal, wind, tidal, solar, hydroelectric and biomass are all renewable. The fossil fuels (coal, oil, natural gas) and nuclear fuel are not. She knows this, as her Section C answer proves, so this is a matter of slowing down on the MCQ.

Spelling to tidy (no marks lost)

"accelaration" for acceleration and "deaccelaration" for deceleration (both used repeatedly), "feul" for fuel, "concid" for coincide, "Triangle" for triangle, "clistance" for distance.

Marked 21 August 2026 from physics/exams/exam-14-ch1-6-9/attempts/2026-08-21/scans.pdf (6 pages). All 24 MCQs and every sub-part of Q.2 and Q.3 were present and attempted; no page or question was missing. Deductions logged: 4 MCQ marks + 2 marks at Q.3(i)(b) = 6, and $43 - 6 = 37$, which reconciles with the headline score.